

SHELBY COUNTY AP, AL, USA

WMO: 722300

Lat: 33.178N

Lon: 86.782W

Elev: 565

StdP: 14.40

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53864

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	26.9	7.3	8.2	29.6	11.6	10.1	35.2	18.3	47.2	16.7	47.4	4.7	350	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	95.1	73.8	92.7	74.4	90.6	74.4	78.4	87.6	77.6	86.8	76.9	85.6	5.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	139.1	81.6	75.2	134.7	81.0	74.3	131.0	80.4	42.2	88.1	41.3	86.7	40.5	85.9	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.7	11.2	17.3	97.3	4.7	3.7	13.9	100.0	11.2	102.2	8.5	104.3	5.1	107.0
			15.4	80.2	4.5	1.1	12.1	81.0	9.5	81.6	6.9	82.3	3.6	83.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	45.1	49.6	56.5	64.1	71.5	78.2	80.6	80.5	75.7	65.0	54.6	48.1
	DBStd	14.50	10.20	9.27	8.66	6.77	5.66	3.43	2.91	3.45	5.30	7.85	8.29	9.06
	HDD50	575	220	113	39	3	0	0	0	0	0	3	46	151
	HDD65	2404	618	435	282	95	17	0	0	0	4	100	325	528
	CDD50	5761	68	102	242	427	668	846	948	946	772	467	184	91
	CDD65	2116	2	5	20	69	220	396	483	481	325	99	13	3
	CDH74	18550	2	16	149	560	1781	3491	4564	4478	2666	771	67	5
	CDH80	7507	0	0	16	89	566	1476	2036	2063	1058	195	8	0
Wind	WSAvg	4.0	5.0	5.2	5.2	4.8	3.7	3.1	3.0	2.7	3.0	3.5	4.0	4.6
Precipitation	PrecAvg	56.0	5.3	5.5	5.9	4.8	3.7	4.6	5.2	4.1	4.1	3.1	4.2	5.4
	PrecMax	70.9	9.2	9.3	9.9	9.6	9.5	9.4	11.1	9.2	10.2	8.9	10.2	10.4
	PrecMin	31.3	2.6	1.9	1.4	1.5	0.4	1.9	1.9	0.0	0.3	0.2	1.0	2.5
	PrecStd	9.1	1.9	2.0	2.2	1.9	2.0	2.2	2.4	2.0	2.8	2.1	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	76.5	82.2	85.2	90.8	96.3	97.3	99.5	95.6	88.3	80.8	73.9
		MCWB	63.6	65.4	64.0	67.6	71.7	72.4	75.1	74.3	71.7	70.2	66.5	66.3
	2%	DB	69.1	72.3	78.1	82.5	88.0	92.7	93.5	95.5	92.0	84.7	75.1	70.2
		MCWB	61.5	62.3	63.0	66.6	70.9	73.4	75.9	74.4	72.4	68.7	63.3	63.8
	5%	DB	65.1	68.7	74.7	80.4	85.6	90.4	91.4	92.7	89.2	81.8	72.2	66.6
		MCWB	60.0	60.4	61.5	65.5	70.0	73.5	75.8	74.7	72.5	67.0	62.5	61.8
	10%	DB	61.8	65.1	71.2	77.5	83.3	88.1	89.7	90.1	86.4	78.7	69.1	63.3
		MCWB	57.2	57.5	60.0	63.7	69.3	73.4	75.6	75.0	72.3	66.3	61.6	58.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	67.7	68.4	72.2	75.8	78.3	80.2	79.5	77.3	75.7	71.0	69.2
		MCDB	70.2	72.8	73.9	80.4	83.9	87.0	89.2	89.0	85.8	80.0	74.9	72.0
	2%	WB	64.0	65.5	66.5	69.8	74.0	77.2	78.6	78.3	76.1	73.2	68.6	65.7
		MCDB	66.8	69.9	72.7	77.2	82.2	86.3	87.8	87.7	84.4	79.8	71.7	68.2
	5%	WB	61.5	63.2	64.5	68.3	72.7	76.3	77.9	77.4	75.2	71.4	65.7	63.0
		MCDB	64.2	66.9	71.2	75.6	81.1	85.0	87.0	86.7	83.1	77.7	70.3	65.6
	10%	WB	57.9	59.7	62.4	66.8	71.5	75.3	77.0	76.7	74.2	69.3	62.7	59.4
		MCDB	61.0	63.5	68.9	73.8	79.8	83.5	85.8	85.3	81.6	74.9	67.4	62.8
Mean Daily Temperature Range	5% DB	MDBR	19.1	20.0	20.8	21.4	19.9	18.5	18.1	18.6	19.3	21.2	21.8	18.6
		MCDBR	20.8	22.8	23.6	23.7	22.3	22.3	20.6	22.8	22.5	24.4	24.0	21.1
		MCWBR	14.4	14.2	11.5	10.2	8.2	7.4	6.5	6.7	7.2	10.4	13.8	15.2
	5% WB	MCDBR	17.1	18.9	18.7	18.8	17.8	17.9	18.0	18.0	17.1	17.0	17.4	17.6
		MCWBR	14.3	14.7	11.3	10.1	8.0	7.5	6.8	6.5	6.8	9.0	12.5	15.1
Clear-Sky Solar Irradiance	taub	0.319	0.327	0.359	0.390	0.451	0.475	0.526	0.524	0.431	0.368	0.342	0.322	
	taud	2.530	2.504	2.430	2.327	2.198	2.172	2.049	2.045	2.296	2.461	2.472	2.535	
	Ebn at Noon	286	294	291	285	267	260	246	244	265	276	273	278	
	Edh at Noon	26	30	35	40	46	48	53	53	39	31	28	25	
All-Sky Solar Radiation	RadAvg	848	1017	1383	1757	1923	1958	1894	1750	1541	1290	958	723	
	RadStd	63	132	127	126	147	149	125	99	136	163	99	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+187
Regional (46 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+193

Nomenclature: See separate page